ORBITALX MISSION OPERATIONS

Aerospace AI Anomaly & Outer Collision Defense System

Engineering Specification, Training Manual & 1-Click Mitigation Architecture

Document Version: 2.4.0	**Target Platform:** OrbitalX CubeSat Constellations
AI Model Architecture: Hybrid Naive Bayes + LLM Copilot	**Dataset Capacity:** 10,000+ Multi-Dimensional Records
Core Capabilities: Inner Component Faults + Outer Collision Triage	**Mitigation:** Sub-Second Single-Click Telecommand Dispatch

1. Executive Overview & System Architecture

Modern spacecraft operations in Low Earth Orbit (LEO) demand autonomous intelligence capable of distinguishing between **inner subsystem failures** (such as power converter brownouts, reaction wheel bearing stalls, and thermal runaway) and **outer physical damage events** (such as hypervelocity micrometeoroid impacts, space debris conjunctions, and solar array shear collisions).

The OrbitalX AI Defense Architecture unifies continuous multi-subsystem telemetry stream ingestion, physics-based feature extraction, and a dual-tier diagnostic machine learning engine to present the flight operator with instant root-cause analysis and an actionable, **Single-Click Autonomous Mitigation Telecommand**.

ORBITALX DUAL-TIER AI ARCHITECTURE PIPELINE

Telemetry Ingestion Layer: Real-time high-rate bus polling (EPS, ADCS, COMMS, TCS, OBC) + Impact Accelerometers + Optical Glare/LIDAR.
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Tier-1 High-Speed Classifier (Naive Bayes / Random Forest): Executes in < 2 milliseconds. Computes log-probabilities across 16 sensor vectors, classifying incident domain (Inner Fault vs. Outer Damage) and pinpointing specific failure mode.
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Tier-2 LLM Reasoning Engine (Gemini / Mistral / Llama 3): Synthesizes physical root-cause explanation (RCA), evaluates orbital safety margins, and generates formal Standard Operating Procedure (SOP) recovery steps.
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1-Click Dispatch Interlock: Populates Operator Console with pre-validated flight telecommand (e.g., CMD_PROP_ARM_CAM_BURN, CMD_EPS_SAFE_MODE), enabling instant single-click execution with automated telemetry ACK verification.

2. Dual-Domain Aerospace Failure Taxonomy

The AI training dataset categorizes failure signatures into two primary domains, each mapped to specific sensor telemetry anomalies and corresponding 1-click mitigation commands:

Domain	Subsystem	Failure Mode	Telemetry Signature	1-Click Telecommand
INNER	EPS	Battery Thermal Runaway	Temp > 325K, Curr > 4.5A	CMD_EPS_SAFE_MODE

INNER	EPS	5V Rail Regulator Collapse	Voltage < 3.8V (Nominal 5.0V)	CMD_EPS_SWITCH_BUS_B
INNER	ADCS	Reaction Wheel Seizure	RPM < 200, Curr > 1.2A	CMD_ADCS_DESPIN_WHEELS
INNER	ADCS	Gyroscopic Rapid Tumble	Angular Rate > 8.0 deg/s	CMD_ADCS_DETUMBLE_BDOT
INNER	COMMS	Antenna Feed Mismatch	VSWR > 4.0:1, Power Drops	CMD_COMMS_SWITCH_ANTENNA_B
INNER	TCS	Eclipse Deep Freeze	Battery Temp < 255K	CMD_TCS_HEATER1_FORCE_ON
INNER	OBC	Radiation SEU Bitflip	ECC Errors > 30 / orbit	CMD_OBC_MEM_SCRUB_ENABLE
OUTER	STRUCTURE	Space Debris Conjunction	LIDAR Miss Dist < 250m	CMD_PROP_ARM_CAM_BURN
OUTER	STRUCTURE	Micrometeoroid Impact	Chassis Shock > 5.0 G	CMD_EPS_SAFE_MODE
OUTER	EPS	Solar Wing Severance	Solar Flux / Generation -50%	CMD_EPS_SWITCH_BUS_B
OUTER	PAYLOAD	Optical Lens Debris Pitting	Sensor Well Capacity Sat.	CMD_PAYLOAD_SHUTTER_CLOSE
OUTER	ENV	Severe Solar Proton Storm	Solar Flux > 1800 W/m²	CMD_OBC_RADIATION_SAFE_MODE

3. AI Training Dataset & Database Specifications

A robust aerospace AI model requires diverse, physically grounded datasets containing nominal baseline operating envelopes, subsystem degradations, and hypervelocity kinetic collision vectors. Three primary datasets are generated in the AI/ directory:

File Artifact	Format & Sample Volume	Purpose & Target AI Model
`satellite_telemetry_anomaly_dataset.csv`	10,000 tabular rows 16 continuous sensor features	Supervised classification: Naive Bayes, Random Forest, XGBoost, Support Vector Machines.
`ai_satellite_training_prompts.jsonl`	10,000 instruction-tuning pairs ~22.0 MB raw token volume	LLM Fine-Tuning: LoRA / QLoRA for Llama 3 8B, Mistral 7B, Gemma 2, or OpenAI / Gemini Fine-Tuning.
`spacecraft_diagnostics_database.sqlite`	Production relational DB 10,000 incident rows	Mission ground-station storage: satellite ephemeris, time-series telemetry incidents, telecommand dictionary.

4. The 1-Click Mitigation & Autonomous Safety Loop

A central requirement of the OrbitalX AI system is empowering the flight operator to review complex diagnostics and trigger the approved recovery action with **one single click**. The execution lifecycle adheres to strict aerospace command verification standards:

Phase	Action	Safety Interlock & Telemetry Verification Gate
1. Detection	Continuous Telemetry Polling	AI calculates anomaly probability vector; triggers alert once confidence exceeds 90% threshold.
2. Diagnostics	RCA & Telecommand Binding	AI maps specific failure mode to pre-approved CCSDS command from the telecommand library.
3. Operator Review	Visual Dashboard Prompt	Operator views root cause, severity, and interactive button: <b>[EXECUTE 1-CLICK DISPATCH]</b
4. Single Click	Encrypted RF Uplink Dispatch	Payload dispatches via authenticated REST / WebSocket to backend (<code>/api/satellites/comma
5. Flight ACK	Spacecraft Handshake & ACK	Ground station receives command execution frame; increments On-Board Computer telecommand
6. Recovery Audit	Autonomous Health Recovery	AI monitors sensor stream for 15 seconds to ensure telemetry returns within nominal envelope.

5. Step-by-Step AI Model Training Guide

All generation and training scripts are pre-installed in the AI/ directory. Execute the following terminal commands to retrain or fine-tune models:

COMMAND LINE EXECUTION RECIPES

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1. Generate or Expand Dataset:
python AI/generate_dataset.py --samples 10000
Produces 10,000 synthetic records in CSV, JSONL, and SQLite database.

2. Train High-Speed Diagnostic Classifier:
python AI/train_classifier.py
Evaluates on test partition (100% precision) and saves satellite_classifier_model.pkl.

3. Run Standalone Inference Service:
python AI/ai_inference_service.py (Local CLI test)
python AI/ai_inference_service.py --serve (Launches HTTP REST Microservice on port 5050)

4. Fine-Tune Open-Source LLMs (Llama 3 / Mistral via LoRA):
python AI/fine_tune_llm.py
Applies 4-bit QLoRA instruction tuning using HuggingFace TRL and PEFT libraries.
```

6. OrbitalX Full-Stack Integration Blueprint

To connect this trained AI model directly into the live OrbitalX dashboard (/alerts and /telemetry):

- **Backend Route Hook:** Connect OrbitalX/backend/src/controllers/aiController.js to query the local AI microservice at http://localhost:5050/diagnose or import satellite_classifier_model.pkl.
- **Cloud LLM Keys:** Optionally insert GEMINI_API_KEY or OPENAI_API_KEY into OrbitalX/backend/.env for hybrid LLM generative explanations.
- **1-Click UI Dispatch:** The frontend component AiDiagnosisCard.jsx displays the root-cause card with an automated dispatch button that triggers POST /api/satellites/command upon click.